Army Vantage

(Palantir Foundry-based)

Contents

Overview

Army Vantage (Palantir Foundry-based)

1. Vantage Basics

- Open database / source data — You don’t “open” a traditional database. Instead:
  - Data is ingested via Data Connection (automated pipelines from authoritative Army systems).
  - You explore data through the Ontology (real-world objects like Units, Soldiers, Equipment) or Object Explorer / Force Explorer.
  - Use Contour for no-code / low-code analysis on large datasets.
- Run a query —
  - Point-and-click in Contour, or use Fusion (spreadsheet-like).
  - For advanced users: write code in Python/R or use built-in query tools.
- Export to file — Easy export to CSV, Excel, etc., from any view, table, or report.

2. Automation & Power Automate / Office 365

- Vantage has strong native automation via Pipelines and scheduled Data Connections.
- Direct Power Automate integration is limited, but you can trigger actions via APIs or export data that Power Automate can pick up.
- Strong Microsoft 365 ties overall (see below).

3. Create & Share Dashboards

- Use Workshop → point-and-click app/dashboard builder. Embed charts, object sets, tables, maps, etc.
- Use Reports for polished, document-style outputs with embedded visuals.
- Sharing is role-based and very secure (common in DoD environments).
- Yes — user management can integrate with MS Entra ID / Office 365 Active Directory (especially in Army environments). Many units use this for authentication.

4. Use with Power BI Desktop

- Direct connector exists — Palantir provides an official Power BI Connector (ODBC + Power Query) and a REST-based option.
- You can connect directly to Foundry datasets/objects and pull live or refreshed data into Power BI.
- Automation for PBI modeling —
  - Use scheduled pipelines in Vantage to prepare data.
  - Power BI Gateway + Foundry connector for automated refreshes in the Power BI Service.

  - Expanded Vantage-to-Power BI connector is widely used in the Army (hundreds of thousands of connections per month).
- Report builder — Vantage has its own strong report/dashboard tools (Workshop + Reports). Many users build final products in Vantage Workshop and/or export to Power BI depending on audience needs. They complement each other rather than compete.

Quick Start Tip: Most Army users begin in Contour or Object Explorer for exploration, then move to Workshop for dashboards or export to Power BI for broader distribution.

Quick user manual

▯ Army Vantage – Basic Steps

1) Open the Platform

1. Navigate to:
   👉 https://vantage.army.mil
2. Log in with CAC (Army network required)
3. Use the **left navigation panel** to access tools (Projects, Data, Reports, etc.)
   [financialf...e.army.mil]

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2) Open a Query / Analysis Tool

You don't typically open a "SQL editor" like a database — instead you use:

👉 **Contour (analysis tool)** or **Workshop (apps)**

- From left menu → select **Contour** (point-and-click analysis) [financialf...e.army.mil]

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3) Run a Basic "Query"

(Vantage is mostly no-code, but conceptually this is your query step)

Example A – View dataset (like "SELECT *")

1. Open **Contour**
2. Select a dataset / Object (e.g., Units, Personnel, Logistics)
3. Apply filters or choose columns

✅ This = your "basic query result table"

Example B – Filter data (pseudo-query)

Use filters like:

Unit = "XYZ"

Status = "Active"

Date > Last 30 days

✅ This behaves like a SQL `WHERE` clause

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4) View Data Sources / Available Data

1. Go to **Object / Force Explorer**
2. Browse:
   - Objects (datasets)
   - Properties (columns)
3. Explore relationships between datasets [financialf...e.army.mil]

✅ This = your "list databases/tables"

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5) Export Data

1. From a table or report:
   - Click **Export / Download**
2. Choose format:
   - CSV
   - Excel

✅ Data is downloaded locally or shared within Vantage

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6) Create & Share Report / Dashboard

1. Go to **Reports**
2. Add:
   - tables
   - charts
3. Save report
4. Share with users or groups

✅ Reports = your "dashboard layer" [financialf...e.army.mil]

⚠️ Important Differences (vs SQL tools)

- No direct "run `SELECT version()`" command
- No standard SQL console exposed to most users
- Everything is:
  - object-based
  - filter-driven
  - UI + low-code

🧠 Quick Translation (SQL → Vantage)

SQL action	Vantage equivalent
`SELECT * FROM table`	Open object in Contour
`WHERE` clause	Filters
`SHOW DATABASES`	Object Explorer
Export query results	Export button
BI dashboard	Reports

✅ One-Line Workflow

Login → Open Contour → Select dataset → Apply filters → View results → Export or build report

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Power BI + Vantage step-by-step

✅ 0) What this setup uses

- Palantir Vantage (data platform)
- Power BI Desktop
- Foundry / Vantage connector (behind the scenes)

👉 There is an official **Power BI connector for Palantir Foundry** (what Vantage runs on) [learn.microsoft.com]

🧭 1) Prepare Data in Vantage

1. Log into Vantage → https://vantage.army.mil
2. Navigate to:
   - **Contour** (analysis) OR
   - **Ontology / Objects**
3. Build your dataset:
   - Select object (dataset)
   - Apply filters
   - Choose fields/columns
4. Save it as:
   - **Data Product** OR
   - Saved dataset/report

✅ This is your "Power BI data source"

🔌 2) Connect Power BI Desktop

In Power BI Desktop:

1. Go to **Home** → **Get Data**
2. Search for: 👉 **Palantir Foundry**
3. Enter:
   - Base URL (your Vantage / Foundry instance)
   - Dataset ID (if required)
4. Choose:
   - **Import** (best for modeling)
   - OR **DirectQuery** (live connection) [learn.microsoft.com]
5. Sign in:
   - Organizational account (CAC-backed / OAuth)

✅ This establishes the connection

📊 3) Select & Load Data

1. In Navigator:
   - Browse available datasets
2. Select your dataset(s)
3. Click:
   - **Load** → quick start
   - OR **Transform Data** → Power Query

✅ Data now in Power BI model

🗃 4) Build Your Model (Power BI)

1. Define:
   - relationships
   - measures (DAX)
2. Create visuals:
   - tables
   - charts
   - KPIs

✅ This is your reporting layer

🔄 5) Automate Refresh

Option A — Import Mode

1. Publish to Power BI Service
2. Configure:
   - Scheduled refresh
   - Gateway (if needed)

Option B — DirectQuery

- No scheduled refresh required
- Data stays live in Vantage

✅ Used for near-real-time scenarios

📤 6) Share Reports

1. Publish to:
   - Power BI Workspace
2. Share:
   - dashboard
   - app
   - report links

✅ Works with M365 permissions

🔐 7) Authentication & Access

- Uses:
  - OAuth / Organizational login [learn.microsoft.com]
- Controlled by:
  - Vantage permissions
  - Power BI workspace roles

🧠 Important Notes (from real usage)

Integration Strength

- Army environments actively use this combo:
  - **Vantage + Power BI interoperability is common and high-volume** [engineering.fyi]

Data Flow Pattern

1. Vantage (data pipelines + objects)
2. ↓
3. Saved dataset / data product
4. ↓
5. Power BI (Import or DirectQuery)
6. ↓
7. Dashboards / Apps
8. ``

9.

What runs where?

Function	Tool
Data prep / pipelines	Vantage
Data storage / logic	Vantage
Advanced modeling (DAX)	Power BI
Visualization	Power BI

Quick "Do This First" Checklist

1. ✅ Build dataset in Vantage (Contour)
2. ✅ Save as reusable dataset
3. ✅ Connect via Power BI → Foundry connector
4. ✅ Load + model
5. ✅ Publish + schedule refresh

Vantage → Power BI refresh → SharePoint export → Teams alert

🔄 End-to-End Flow (High-Level)

1. Vantage dataset updated
2. ↓
3. Power BI dataset refresh
4. ↓
5. Export / snapshot to SharePoint
6. ↓
7. Teams notification (with link)

OPTION A (Best Practice): Trigger from Power BI

🧭 Flow Name:

"Vantage Dataset Refresh → Notify & Publish"

✅ 1) Trigger: Power BI Dataset Refresh Complete

In Power Automate:

- Trigger: 👉 **Power BI – When a dataset refresh completes**
- Configure:
  - Workspace
  - Dataset (connected to Vantage)

✅ 2) Condition: Check Success

Add a condition:

Status = "Completed"

If failed → send failure alert (optional)

✅ 3) Export Report Snapshot

Option 1 (Native)

- Action: 👉 **Power BI – Export to File for Report**
- Format:
  - PDF (best for distribution)
  - PPTX (if needed)

Option 2 (Data-level export)

- Use:
  - Power BI REST API
  - OR Paginated Reports (if available)

✅ 4) Save to SharePoint

- Action: 👉 **SharePoint – Create File**
- Location:
  - Document Library (e.g., `/Shared Reports/Vantage/`)
- File naming pattern:

Vantage_Report_@{utcNow('yyyy-MM-dd_HHmm')}.pdf

✅ 5) Post to Teams

- Action: 👉 **Microsoft Teams – Post message in channel**

Example message:

✅ Vantage dataset refresh completed

📊 Report updated: [SharePoint link]

🕒 Time: @{utcNow()}

Dataset: Logistics Overview

OPTION B: Add Data Export Layer (CSV / Table)

If you want **raw data export (more like your earlier pattern)**:

✅ Insert Step After Refresh

Use:

👉 **Run a query against a dataset (Power BI)**

- DAX Query Example:

DAX
EVALUATE 'LogisticsTable'

✅ Convert to File

- Action: 👉 Create CSV table

✅ Save to SharePoint

Same step as above:

- Output:

Vantage_Data_YYYYMMDD.csv

OPTION C: Fully Automated (Scheduled Pull)

If Vantage updates aren't event-driven:

Trigger:

👉 **Recurrence (e.g., every hour)**

Then:

1. Trigger Power BI dataset refresh
2. Wait (delay 2–5 min)
3. Continue export + Teams steps

🔐 Permissions Model (Important)

Layer	Controlled by
Vantage data	Palantir access
Power BI dataset	Workspace roles
SharePoint	Site permissions
Teams	Channel membership

🧠 Real-World Pattern (What most teams do)

"Gold Standard Flow"

Vantage (pipeline builds dataset)

↓

Power BI (semantic model)

↓

Power Automate:

- refresh dataset

- export report

- save snapshot to SharePoint

- notify Teams

💡 Pro Tips (from experience)

✅ Use folders like:

/Shared Documents/Vantage/

/Reports/

/Data Extracts/

✅ Add version tracking

- Include timestamp in file name
- OR overwrite "latest" + keep archive

✅ Add deep link back to Power BI

Include in Teams message:

🔗 Live dashboard: https://app.powerbi.com/...

✅ Failure branch

Add parallel condition:

- If refresh fails → Teams alert:

❌ Vantage dataset refresh FAILED

⚡ Minimal Version (if you want super lean)

Trigger: Dataset refresh complete

→ Export report (PDF)

→ Save to SharePoint

→ Post Teams message

📦 1) Power Automate JSON Template (Basic)

⚠️ Note: This is a **minimal working template** you can import and then update connections/IDs.

JSON

```json
{
"definition": {
"$schema": "https://schema.management.azure.com/providers/Microsoft.Logic/schemas/2019-05-01/workflowdefinition.json#",
"actions": {
"Export_Report": {
"type": "ApiConnection",
"inputs": {
"host": {
"connection": {
"name": "@parameters('$connections')['powerbi']['connectionId']"
}
},
"method": "post",
"path": "/v1.0/myorg/groups/@{encodeURIComponent('WORKSPACE_ID')}/reports/@{encodeURIComponent('REPORT_ID')}/ExportTo",
"body": {
"format": "PDF"
```

}

}

},

"Create_File_in_SharePoint": {

"runAfter": {

"Export_Report": [

"Succeeded"

]

},

"type": "ApiConnection",

"inputs": {

"host": {

"connection": {

"name": "@parameters('$connections')['sharepointonline']['connectionId']"

}

},

"method": "post",

"path": "/datasets/default/files",

"body": {

"folderPath": "/Shared Documents/Vantage/Reports",

"name": "Vantage_Report_@{utcNow('yyyy-MM-dd_HHmm')}.pdf",

"fileContent": "@body('Export_Report')"

}

}

},

"Post_Teams_Message": {

"runAfter": {

"Create_File_in_SharePoint": [

"Succeeded"

]

},

"type": "ApiConnection",

"inputs": {

"host": {

"connection": {

"name": "@parameters('$connections')['teams']['connectionId']"

}

},

"method": "post",

"path": "/v3/conversations/CHANNEL_ID/messages",

"body": {

"content": "✅ Vantage dataset refresh completed\r\n📊 Report saved to SharePoint\r\n Time: @{utcNow()}"

}

}

}

},

"triggers": {

"When_dataset_refresh_completes": {

"type": "ApiConnectionWebhook",

"inputs": {

"host": {

"connection": {

"name": "@parameters('$connections')['powerbi']['connectionId']"

}

},

"path": "/datasets/onrefreshcomplete",

"method": "get"

}

}

}

},

"parameters": {

"$connections": {

"value": {

"powerbi": {

"connectionId": "/providers/Microsoft.PowerApps/apis/shared_powerbi",

"connectionName": "shared_powerbi",

"id": "/providers/Microsoft.PowerApps/apis/shared_powerbi"

},

"sharepointonline": {

"connectionId": "/providers/Microsoft.PowerApps/apis/shared_sharepointonline",

"connectionName": "shared_sharepointonline",

"id": "/providers/Microsoft.PowerApps/apis/shared_sharepointonline"

},

"teams": {

"connectionId": "/providers/Microsoft.PowerApps/apis/shared_teams",

"connectionName": "shared_teams",

"id": "/providers/Microsoft.PowerApps/apis/shared_teams"

}

}

}

}

}

2) What You MUST Customize

Before it works, replace these values:

Field	Replace With
`WORKSPACE_ID`	Power BI workspace ID
`REPORT_ID`	Power BI report ID
`CHANNEL_ID`	Teams channel ID
SharePoint path	Your document library path

3) Import Steps (Power Automate)

✅ Method 1 (Recommended – Solutions Import)

1. Go to: 👉 https://make.powerautomate.com
2. Click: **Solutions → Import solution**
3. Upload your `.zip` (wrap JSON in a solution if needed)
4. Map connections:
    - Power BI
    - SharePoint
    - Teams
5. Click **Import**

✅ Method 2 (Quick Manual Setup)

Since raw JSON import is limited in UI:

1. Go to **My Flows**
2. Click **New → Automated cloud flow**
3. Recreate using:
    - Trigger: **Power BI – When a dataset refresh completes**
    - Add actions in same order:
        1. Export report
        2. Create SharePoint file

3. Post Teams message

👉 Then use JSON above as reference for fields

🧪 4) Test the Flow

1. Trigger a dataset refresh in Power BI
2. Wait for completion
3. Verify:
   - ✅ PDF in SharePoint
   - ✅ Teams message posted

⚡ Optional Add-On (Highly Recommended)

Add a success/failure check

Insert condition:

Status = Completed

If FAILED:

Post Teams alert:

❌ Vantage dataset refresh FAILED

🧠 Final “Minimal Working Flow”

Trigger: Dataset refresh complete

→ Export report (PDF)

→ Save to SharePoint

→ Post Teams message

Embedding a Power BI Report in a Teams Site (Army Vantage Context)

You can embed a Power BI report directly into a Microsoft Teams channel or site so Army Vantage users can view and interact with it without leaving Teams. This works if your Power BI content is in a workspace you have access to and your Teams environment supports the Power BI tab.

Requirements Microsoft Learn

- Teams must have the **Power BI tab** enabled.
- You must have at least a **Viewer role** in the Power BI workspace containing the report.
- You must have permission to view the organizational app in Teams.
- End users must be Teams members with access to the channel/site and have permission to view the Power BI item.

Steps to embed in Teams Microsoft Learn**+1**

1. Open the Teams channel or site where you want the report.
2. Click the **+** icon (Add a tab).
3. Search for and select **Power BI** from the list of available apps.
4. Choose how to add the report:
    - **Paste a link**: Enter a supported Power BI link (e.g., a report or dashboard link) and select **Add tab**.
    - **Browse workspaces**: Select the workspace containing your Army Vantage report, then choose the specific report/app.
5. Save the tab. The report will now appear as a clickable tab in the channel/site.

Tips for Army Vantage integration www.financialfrontline.army.mil**+1**

- Ensure the Power BI report is published in a workspace accessible to Army Vantage users.
- If embedding via a link, make sure the link includes the correct permissions so users can view it.
- Army Vantage is best viewed in **Chrome or Firefox** on the DoD network.
- You can also share the report directly in a Teams conversation if you don't want to add it as a tab.

Resetting or changing the embedded item Microsoft Learn

- Right-click the tab name, select **Settings**, then **Reset tab** to reconfigure it.
- You can then browse workspaces or paste a new link to update the content.

By following these steps, you can keep Army Vantage analytics embedded in Teams for seamless collaboration, ensuring users can access live data alongside conversations.

Forecasting with Line-Visual

Understanding Seasonality in Power BI Forecasting

In **Power BI forecasting**, *seasonality* refers to the **predictable, repeating patterns** in your time-series data that occur at fixed intervals — such as monthly, weekly, or yearly cycles www.thebricks.com**+1**. These patterns are driven by factors like the time of year, day of the week, or business quarters, and they are distinct from random noise or long-term trends.

What Seasonality Means in Power BI

When you use the **Forecast** feature in Power BI (available in the Analytics pane of a line chart), the tool applies a statistical model (often Exponential Smoothing with Trend and Seasonality — ETS) to your historical data www.thebricks.com. This model separates:

- **Trend** – the overall long-term direction.
- **Seasonality** – the repeating cyclical component.
- **Random noise** – irregular variations.

The *seasonality* parameter tells Power BI how many data points in your time series represent one full cycle of the seasonal pattern Microsoft Fabric Community.
For example:

- If your data is monthly and you expect a yearly cycle, you set **seasonality = 12**.
- If your data is weekly and you expect a weekly cycle, you set **seasonality = 7**.

How to Choose the Seasonality Value

- **Match your data frequency** to the cycle length.
  Example: 12 months of data → seasonality = 12 Microsoft Fabric Community.
- If you set a value that doesn't match your actual cycle, Power BI may not capture the true seasonal pattern, leading to less accurate forecasts.

- If you set a value that's too small (e.g., 11 for a yearly cycle), Power BI may show upper/lower bounds but may not align with your real-world pattern Microsoft Fabric Community.

Why It Matters

Ignoring seasonality can cause forecasts to be misleading:

- Mistaking a seasonal dip for a permanent decline.
- Being unprepared for a seasonal surge, leading to stockouts or missed sales www.thebricks.com.

Practical Example

A ski resort's occupancy peaks from December to March — a **12-month seasonal cycle**. In Power BI, you'd set seasonality = 12 to capture this pattern, ensuring the forecast reflects the expected rise and fall over the year www.thebricks.com.

In short: In Power BI forecasting, seasonality is the number of data points in one full repeating cycle, and setting it correctly is key to capturing predictable patterns and improving forecast accuracy.

To analyze Vantage data in Power Query or Power BI

Use the **Palantir Vantage connector** along with proper authentication and, in some cases, an ODBC driver Microsoft+1.

Prerequisites

1. **Access to Army Vantage**: You must have an Army CAC and appropriate permissions to access Vantage datasets palantir.com.
2. **Power BI Desktop**: Ensure you have the latest version of Power BI Desktop installed.
3. **ODBC Driver**: Download and install the Palantir Foundry/Vantage ODBC driver if required for your environment palantir.com+1.
4. **Authentication**: OAuth authentication is recommended. Your Vantage administrators must enable the Power BI third-party application in the Vantage Control Panel Microsoft.

Steps to Connect

1. Open **Power BI Desktop** and go to **Get Data**.
2. Select **Palantir Vantage** (or Foundry) from the list of connectors.
3. In **Connection Settings**, enter the **Base URL** of your Vantage environment (e.g., https://<subdomain>.army-vantage.com/).
4. Optionally, provide a **Dataset RID** and **Branch** if you want to connect to a specific dataset.
5. Choose the **data connectivity mode**: Import or DirectQuery.
6. Select **OK**.
7. Authenticate using **Foundry OAuth** (recommended) or a **token** if OAuth is not available.
8. After signing in, select **Connect**.
9. In the **Navigator**, choose the datasets you want to load or transform in Power Query.

Additional Tips

- If using **Power Query Online**, ensure your on-premises data gateway is configured with a connection to Vantage Microsoft.
- Use the **Transform Data** option in Power Query to shape and clean your data before loading it into Power BI reports.
- The integration supports secure, real-time access to Vantage data, enabling advanced analytics and visualization for Army operations engineering.fyi.
  By following these steps, you can leverage Army Vantage data in Power Query to create interactive dashboards, perform data transformations, and support data-driven decision-making across Army operations.

Power BI Apps/Audience Sharing

Create a Power BI App

1. **Set up the app**: In the workspace, select **Create app**. Provide a name, description, and optional theme color. Add a support site link and contact information if needed.
2. **Add content**: On the Content tab, select **Add content** to include dashboards, reports, or other items from the workspace. Arrange the content in the desired order.
3. **Publish the app**: Once the setup and content are finalized, click **Publish app**. A shareable link will be generated for distribution.

Update a Power BI App

1. Open the workspace: Navigate to the workspace associated with the app. Select the **Edit app** pencil icon.
2. **Make changes**: Modify the app's content, setup, or other settings as needed. Changes remain in the workspace until republished.
3. **Republish the app**: Select **Update app** to apply the changes. Users with access will automatically see the updated version.

Unpublish a Power BI App

- **Unpublish the app**: In the workspace, select **More options (...)** > **Unpublish app**.

Note

Impact of unpublishing: This action removes the app from all users and deletes their customizations, such as bookmarks and comments. The workspace and its contents remain intact.

Audiences

Audiences in Power BI apps allow you to tailor the content visibility and access for different groups of users within your organization. This feature is particularly useful when you need to distribute specific dashboards, reports, or other content to distinct teams or departments while maintaining centralized control over permissions and visibility.

What is an Audience in Power BI Apps?

An audience in Power BI apps is a defined group of users who have access to a specific subset of content within an app. By creating multiple audiences, you can customize which content is visible to each group, ensuring that users only see the data relevant to their roles or responsibilities. For example, you might create separate audiences for sales, marketing, and finance teams, each with access to different reports and dashboards.

Why Use Audiences?

Using audiences simplifies content management and enhances security. Instead of creating multiple apps or workspaces for different teams, you can manage all content within a single app and control visibility at the audience level.

How to Use Audiences

To set up audiences in a Power BI app:

1. **Create an Audience Group**: On the Audience tab during app creation or editing, select **New Audience** and name the group.
2. **Customize Content Visibility**: Use the hide/show icons next to each item in the workspace to determine what content is visible to the audience. Hidden content won't appear in the app for that audience unless explicitly allowed via advanced settings.
3. **Assign Users or Groups**: In the Manage audience access pane, specify the users or groups who should belong to the audience. You can also configure advanced permissions, such as allowing users to share or build content with semantic models.
4. **Publish the App**: Once the audiences and their content visibility are configured, publish the app. Users will see only the content assigned to their respective audience groups.

Reference

1. Army Vantage
2. Palantir Foundry Power Query connector - Power Query | Microsoft Learn
3. Create a Power BI app - Training | Microsoft Learn
4. Course PL-300T00-A: Design and manage analytics solutions using Power BI - Training | Microsoft Learn